

Random Forest

Dataset Description:

Use the Glass dataset and apply the Random forest model.

1. Exploratory Data Analysis (EDA):

Perform exploratory data analysis to understand the structure of the dataset. Check for missing values, outliers, inconsistencies in the data.

2: Data Visualization:

Create visualizations such as histograms, box plots, or pair plots to visualize the distributions and relationships between features.

Analyze any patterns or correlations observed in the data.

3: Data Preprocessing

1. Check for missing values in the dataset and decide on a strategy for handling them. Implement the chosen strategy (e.g., imputation or removal) and explain your reasoning.
2. If there are categorical variables, apply encoding techniques like one-hot encoding to convert them into numerical format.
3. Apply feature scaling techniques such as standardization or normalization to ensure that all features are on a similar scale. Handling the imbalance data.

4: Random Forest Model Implementation

1. Divide the data into train and test split.
2. Implement a Random Forest classifier using Python and a machine learning library like scikit-learn.
3. Train the model on the train dataset. Evaluate the performance on test data using metrics like accuracy, precision, recall, and F1-score.

5: Bagging and Boosting Methods

Apply the Bagging and Boosting methods and compare the results.

Additional Notes:

1. Explain Bagging and Boosting methods. How is it different from each other.
2. Explain how to handle imbalance in the data.